



HeliOTrough collector

Spain's Solar Tres Power Station and California's Sierra SunTower.

In fact, CSP is turning out to be an active innovation space these days. In June 2010, Solar Millennium's HeliOTrough, a parabolic trough solar radiation collector technology, won the innovation award at the international CSP conference held annually in San Francisco. HeliOTrough enables over 20 per cent reduction in the cost of setting up a solar power plant, due to its high optical efficiency compared to older collector technologies.

The HeliOTrough collectors are 192m long and 6.7m wide (compared to their 148m long and 5.7m wide predecessors). The collectors have a non-interrupted mirror surface with a significantly increased aperture, and the concentration ratio has been reduced from 82-times to 76-times as the optical efficiency of the collector is significantly better. The diameter of the heat-collecting element is also larger (89 mm), allowing an increased flow of the heat transfer fluid and thus reducing pumping power parasitic. Power plants can also increase plant efficiency by using molten salt as the heat transfer fluid, switching from synthetic oil to a molten salt mixture.

"Solar thermal (CSP) has seen a lot of companies design and demonstrate technologies based on parabolic troughs and linear Fresnel technology,

which look like the most financially-viable sources of energy on a large scale," says Hari Kiran.

CSP plants, such as power towers, could help turn the Rajasthan deserts into a mega-sized solar source for India. However, the general sentiment is that solar power towers are very expensive to set up. Google's new technology could change this situation.

'Small Hydro': Small is big

Having spoken about large implementations such as power towers, now it is time to acknowledge the power of small sources! China is one of the best performers in terms of renewable energy generation and usage, and you will be surprised to know that it creates most of its energy through 'Small Hydro' sources. These are small hydroelectric power generation plants that produce less than 25 megawatts (MW) of energy. They require minimum civil construction, small reservoirs and barely any disruption to the environment.

The power generated by these sources is then augmented at the power grid, adding up to several gigawatts of energy—more than 50 per cent of the renewable energy produced in China.

India also augmented the number of its small hydro projects last year, bringing the total small hydro capacity to around 2430 MW.

Do-it-yourself green energy devices

As far as renewable energy goes, localised production is supposed to be very beneficial as it reduces the wastage that is incurred during transport. Localisation extends to the household level. You can install solar panels, wind mills and other ways to generate power at your home itself.

In the US, do-it-yourself (DIY) green energy devices are becoming a fad, especially for installation in farm-houses, etc. DIY kits are cheaper and also more satisfying. The cost of building a wind turbine is not more than \$100, while a solar panel costs around \$200 and a magnetic generator \$250. Of these, magnetic generators are believed to be better for home installations as they are not dependent on the weather.

Solid-oxide fuel cells all set to power homes

Speaking of energy production at home, enthusiasts who follow energy-related news are sure to have noticed the hype about the solid-oxide fuel cell (SOFC) device demonstrated by Bloom Energy earlier this year. The localised, clean energy production device has been booked by several homes and offices, including eBay and Google. Other start-ups are also launching such fuel cells, which are expected to revolutionise localised energy creation.

Experts claim that such SOFCs are much more practical than hydrogen fuel cells used in some prototype vehicles. SOFCs are electrochemical conversion devices that produce electricity by oxidising a fuel. These can apparently power an entire household, without the heat and transmission losses associated with regular electricity. Plus, the heat generated while producing the electricity can be streamlined to heat water, etc. So an SOFC device saves and produces electricity!

The SOFC device by Bloom looks compact and simple, and can combine natural gas with oxygen from the air, without cumbersome combustion processes, to generate electricity on a small